Primer-Application — Standard Operating Procedure

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> **Audience:** Maintenance Tech · Line Manager **Applies to:** Line-D, station 3, Primer-Application

1. Purpose

Primer-Application sprays a primer coat onto the washed part to promote adhesion of the topcoat applied at the Paint-Booth.

2. Normal operating envelope

- Ideal cycle time: **25 s**
- Max acceptable cycle time: **33 s**
- Expected reject rate: **≤ 3 %**
- Fault probability per cycle: **0.5 %**
- Input buffer capacity: **5 parts**

3. Idle reasons

`idle_reason`	Meaning here	First check
`Starved`	Chemical-Wash (station 2) is not feeding parts.	Verify Chemical-Wash state.
`Blocked`	Paint-Booth (station 4) cannot accept the primed part.	Verify Paint-Booth state.

4. Faults & corrective actions

4.1 `Nozzle-Clog`

- **Symptoms in telemetry:** `fault_type = "Nozzle-Clog"`. Spray fan asymmetric; thickness rejects rise at Coating-Inspection.
- **Likely root cause:** dried primer in the nozzle from a prolonged idle period; foreign matter in the supply line.
- **Corrective action:**
  1. Flush the nozzle with solvent.

2. If still clogged, remove and clean nozzle in ultrasonic bath.
- **Restart criteria:** spray pattern verified against the test pattern card.

4.2 `Pressure-Drop`

- **Symptoms in telemetry:** `fault_type = "Pressure-Drop"`. Coating thickness drops out of spec.
- **Likely root cause:** supply pump failing or air-assist regulator drift.
- **Corrective action:**
  1. Verify air pressure at the gun.
  2. Inspect supply pump output.
- **Restart criteria:** stable atomization pressure for 5 minutes.

4.3 `Viscosity-Off`

- **Symptoms in telemetry:** `fault_type = "Viscosity-Off"`. Coating thickness highly variable; downstream Coating-Inspection rejects mixed.
- **Likely root cause:** primer batch out of spec, or thinner ratio drifted.
- **Corrective action:**
  1. Sample primer; measure with the viscosity cup.
  2. Adjust thinner to recipe.
  3. Re-validate against the spray test card.
- **Restart criteria:** viscosity within ± 2 s of recipe.

5. Preventive maintenance schedule

- **Daily:** purge gun at shutdown; check viscosity at startup.
- **Weekly:** clean nozzles; check filters.
- **Monthly:** supply pump service.
- **Quarterly:** full atomization-system tune.

6. References

- `Maintenance_Workflow_Overview.pdf`
- `Line-D_02_Chemical-Wash_SOP.pdf`
- `Line-D_04_Paint-Booth_SOP.pdf`
- `Line-D_06_Coating-Inspection_SOP.pdf`